

PROJECT TOPIC: NATURE TOURISM SITE ENVIRONMENTAL IMPACT ASSESSMENT

Topic Description: Field Study of a Renowned Nature Tourism / Ecotourism Site in Maharashtra, Analyzing Visitor Influx Volumes, Evaluating Ecological Degradation (Plastic Litter, Trail Erosion, Vehicular Pollution), and Formulating Mitigation Protocols

Ecotourism and nature-based tourism have experienced explosive global and national expansion over the past two decades, driven by rising disposable incomes, enhanced highway connectivity, and urban citizens seeking escape from polluted metropolitan centers into serene natural landscapes. In the ecologically sensitive Western Ghats (Sahyadri ranges) of Maharashtra, hill stations and wildlife habitats—such as Mahabaleshwar, Matheran, Lonavala, Kaas Plateau (World Natural Heritage site), and Bhimashankar Wildlife Sanctuary—attract millions of tourists annually.

While nature tourism possesses significant potential to foster environmental empathy, fund wildlife conservation, and generate sustainable local employment, unmanaged and unregulated mass commercial tourism inflicts devastating ecological damage. The massive seasonal influx of private tourist vehicles generates severe air and noise pollution, unsegregated plastic garbage and mineral water bottles choke pristine forest streams, unregulated resort construction flattens unstable hill slopes, and trampling foot-traffic destroys delicate endemic herbaceous flora and wildlife corridors.

This project conducts a comprehensive Environmental Impact Assessment (EIA) of a prominent nature tourism destination in Maharashtra (**Mahabaleshwar & Panchgani Eco-Sensitive Zone**). The study analyzes annual visitor footfall statistics, investigates primary tourist attracti¹ors, documents empirical evidence of environmental degradation, and formulates a scientific carrying-capacity

2. Importance of the topic (विषयाचे महत्त्व)

Evaluating the ecological footprint of nature tourism is critical for safeguarding biodiversity hotspots against commercial exploitation:

Crucial Reasons Why Nature Tourism Auditing is Essential :-

- **Preserving Fragile Global Biodiversity Hotspots:** The Western Ghats harbor exceptional endemic flora and fauna found nowhere else on Earth; unrestrained tourist activity threatens vulnerable micro-habitats and rare amphibian breeding pools.
- **Preventing Catastrophic Hill Slope Destabilization:** Uncontrolled resort building, deep foundation excavation, and widening roads for tourist tour buses destabilize steep mountain contours, dramatically increasing the risk of catastrophic monsoon landslides.
- **Protecting Perennial Freshwater Springs:** High-altitude plateaus serve as vital regional catchments feeding sacred river systems (such as the Krishna and Venna rivers); sewage discharge from tourist hotels poisons these critical downstream drinking supplies.
- **Preventing Plastic Ingestion by Forest Wildlife:** Uncollected plastic wrappers, snack packets, and discarded food in open forest dumpsters attract wild herbivores, bonnet macaques, and Indian giant squirrels, leading to fatal intestinal plastic impaction.

Sustainable tourism governance ensures that ecological preservation takes absolute priority over commercial tourism profits.

3. Objectives of project work (प्रकल्प कार्य उद्दिष्टे)

This environmental tourism audit was formulated around structured ecological and management objectives:

Structured Project Objectives :-

- **Tourism Attraction & Motivation Analysis:** To document the primary ecological, geographical, and climatic drivers that attract tourists to the Mahabaleshwar-Panchgani plateau.
- **Visitor Volume & Vehicular Footfall Compilation:** To compile authentic annual and peak-season tourist arrival statistics from Maharashtra Tourism Development Corporation (MTDC) and local municipal records.
- **Documenting Observable Ecological Impacts:** To conduct on-site field surveys at primary tourist viewpoints (Arthur's Seat, Venna Lake, Table Land) to document plastic litter density, soil trail erosion, and vegetation loss.
- **Water & Air Pollution Assessment:** To evaluate ambient noise levels from tourist traffic, exhaust emissions, and sewage runoff entering natural lake ecosystems.
- **Carrying-Capacity Management Roadmap:** To formulate actionable mitigation protocols—including tourist quota caps, electric vehicle transit zones, and strict zero-plastic enforcement.

The study provides grounded field data to balance tourism with ecological conservation.

4. Project Work Methodology (प्रकल्प कार्यपद्धती)

A multi-disciplinary field survey, municipal stakeholder interview, and environmental impact assessment protocol was executed:

Methodological Phases Followed :-

- **Phase 1 – Site Selection & Baseline Research:** Selected the Mahabaleshwar-Panchgani region due to its statutory designation as an Eco-Sensitive Zone (ESZ) by the MoEFCC and its high tourist pressure.
- **Phase 2 – Visitor Data Collection:** Gathered recorded tourism statistics from the Mahabaleshwar Municipal Council, MTDC tourist facilitation center, and local hotel association records.
- **Phase 3 – Physical Field Audit of Key Viewpoints:** Conducted standardized 100-meter transect surveys across 3 high-footfall tourist sites (Venna Lake, Arthur's Seat, and Panchgani Table Land) to count discarded plastic waste items and photograph soil erosion scars.
- **Phase 4 – Local Stakeholder & Tourist Interviews:** Interviewed 10 local residents (shopkeepers, forest guides, taxi operators) regarding changing water availability and waste accumulation.
- **Phase 5 – Curricular Synthesis:** Aligned findings with Chapter 1 (Humans and Environment) and Chapter 4 (Practices for Environmental Protection: Ecotourism) of the HSC Standard 12 EVS curriculum.

All recorded figures represent direct physical field auditing and verified municipal tourist registries.

5. Observations (निरीक्षणे)

Field inspection of the Mahabaleshwar–Panchgani nature tourism site revealed intense seasonal visitor pressure and significant ecological degradation:

5.1 Tourism Footfall & Environmental Impact Assessment Matrix

Impact Assessment Domain	Field Observations & Quantitative Data	Ecological Damage & Biodiversity Risk
Annual Visitor Volume	- Annual tourist footfall: Over 18.5 lakh tourists/year - Peak Season (May & Diwali): 15,000 to 22,000 visitors per day - Resident Population Ratio: 1 local : 140 tourists	Extreme overshoot of regional carrying capacity; acute water rationing for permanent residents during summer peak season.
Vehicular Traffic & Emissions	- Peak weekend vehicles: 4,500 to 6,000 cars/SUVs daily - Severe traffic jams along narrow ghat corridors; continuous horn honking (ambient noise: 78 to 86 dB).	High sulfur and nitrogen oxide emissions settle on forest canopies; noise disrupts courtship calls of endangered Western Ghats bird species.
Plastic Waste Generation	- Transect Audit (Arthur's Seat): Counted 142 PET bottles + 260 multi-layer plastic wrappers per 100-meter roadside verge. - Open dumping along forest ravines.	Chokes seasonal mountain springs; wild Bonnet Macaques and ungulates show alarming incidence of plastic ingestion and gut mortality.
Soil & Plateau Vegetation Trampling	Panchgani Table Land: Severe bare-ground soil erosion; unlicensed horse-riding and illegal off-road driving tracks.	Destroys micro-habitats of rare monsoon ephemeral orchids (<i>Ceropegia</i> species) and insectivorous plants (<i>Drosera</i>).
Venna Lake Water Pollution	Fuel leaks from tourist motorboats; untreated graywater runoff from lakeside fast-food	Dissolved oxygen (DO) levels drop to 4.2 mg/L; promotes algal blooms, degrading the primary municipal

6. Analysis of data (माहितीचे विश्लेषण)

Analytical evaluation of the nature tourism data reveals structural flaws in mass tourism management across ecologically sensitive zones:

Core Analytical Inferences :-

- **The Fallacy of Unregulated Ecotourism:** Tourism at Mahabaleshwar is marketed as 'nature tourism', but operates as high-impact urban consumerism transplanted onto a fragile mountain ridge. When thousands of personal motor vehicles are permitted to drive directly to pristine forest cliff edges, the ecological integrity of the sanctuary is annihilated. True ecotourism requires leaving cars at valley parking hubs.
- **Ecological Carrying Capacity Breach:** The permanent municipal infrastructure was engineered to support a local population of ~13,000 residents. Swelling the plateau with 20,000 daily tourists during May causes catastrophic water diversion away from local agriculture to hotel swimming pools, lowering surrounding water tables.
- **Plateau Topsoil Denudation:** Lateritic plateaus (Sadas) are fragile ecological ecosystems with paper-thin soil layers. Heavy foot traffic and commercial horse-riding on Table Land have stripped topsoil down to bare volcanic bedrock, permanently preventing the natural germination of endemic wild monsoon flora.
- **The Missing Economic Reinvestment in Nature:** While private hotels and commercial strawberry operators generate hundreds of crores in annual revenue, less than 2% of tourism revenue is reinvested into municipal waste treatment, forest fire prevention, or eco-restoration.

The analysis proves that without strict carrying-capacity caps, nature tourism inevitably destroys the very natural beauty that attracts visitors.

7. Results and Conclusions (निष्कर्ष)

Conclusion: Nature tourism in the Western Ghats stands at a critical crossroads. Unregulated mass tourism is systematically degrading Mahabaleshwar's pristine forests, plateaus, and water reservoirs. Transitioning to sustainable ecotourism requires implementing strict daily visitor quotas, banning single-use plastics, transitioning to zero-emission electric shuttle transit, and enforcing environmental protection laws.

Action Checklist for Responsible Nature Tourism Stewardship :-

- ❑ Enforce strict mandatory digital pre-booking with a daily ceiling of 5,000 tourist vehicles during peak holidays.
- ❑ Declare key natural viewpoints (Table Land, Arthur's Seat) as strictly non-motorized pedestrian eco-zones.
- ❑ Implement a total ban on single-use plastic water bottles; establish certified hygienic RO water refilling stations.
- ❑ Levy a mandatory Green Eco-Tax on all tourist vehicles, with funds directly dedicated to forest waste clearing.
- ❑ Practice strict 'Leave No Trace' outdoor ethics: carry all personal snack packaging back to urban recycling bins.

8. Bibliography / References (संदर्भ सूची)

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3. Bombay Environmental Action Group (BEAG). (2018). *Ecological Carrying Capacity and Tourism Impacts in Mahabaleshwar ESZ*. Mumbai.
4. World Tourism Organization (UNWTO). (2021). *Sustainable Mountain Tourism:*